

# Expelexia Lab - Data Analysis Report

**File Name:** data.csv  
**Date of Analysis:** 2026-03-26 14:48  
**Prepared For:** Expelexia (AI Analytics Program)  
**Project:** Expelexia Lab

This report provides a comprehensive analysis and AI-generated insights based on the uploaded data.

## Executive Summary

This report summarizes analysis results for data.csv using the same data shown in the website dashboard. The dataset contains 944 records across 10 variables. Main areas needing attention: footfall: 18 anomalies, high variance, tempMode: high variance, AQ: high variance. Observed trend signals: footfall down, tempMode up, AQ down. Recommendations focus on practical monitoring, early detection of unusual changes, and improving reliability of future data collection.

## Data Overview

| Measure             | Value                                                                                                               | What it means                                              |
|---------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Number of records   | 944                                                                                                                 | How many entries were analyzed.                            |
| Number of variables | 10                                                                                                                  | How many fields were evaluated.                            |
| Missing values      | {'footfall': 0, 'tempMode': 0, 'AQ': 0, 'USS': 0, 'CS': 0, 'VOC': 0, 'RP': 0, 'IP': 0, 'Temperature': 0, 'fail': 0} | Missing entries can reduce result reliability if frequent. |

## Website Dashboard Alignment

This table mirrors key trend, focus, and confidence indicators used in the website dashboard view.

| Metric   | Trend | Focus Area                  | Confidence |
|----------|-------|-----------------------------|------------|
| footfall | down  | 18 anomalies, high variance | 0.980      |
| tempMode | up    | high variance               | 1.000      |
| AQ       | down  | high variance               | 1.000      |
| USS      | up    | high variance               | 1.000      |
| CS       | down  | 13 anomalies, high variance | 0.990      |
| VOC      | up    | high variance               | 1.000      |
| RP       | up    | high variance               | 1.000      |
| IP       | down  | high variance               | 1.000      |

| Metric      | Trend  | Focus Area    | Confidence |
|-------------|--------|---------------|------------|
| Temperature | up     | high variance | 1.000      |
| fail        | stable | high variance | 1.000      |

Visual Analysis

The chart below shows how key values compare. It helps identify increases, decreases, and unusual patterns over time.

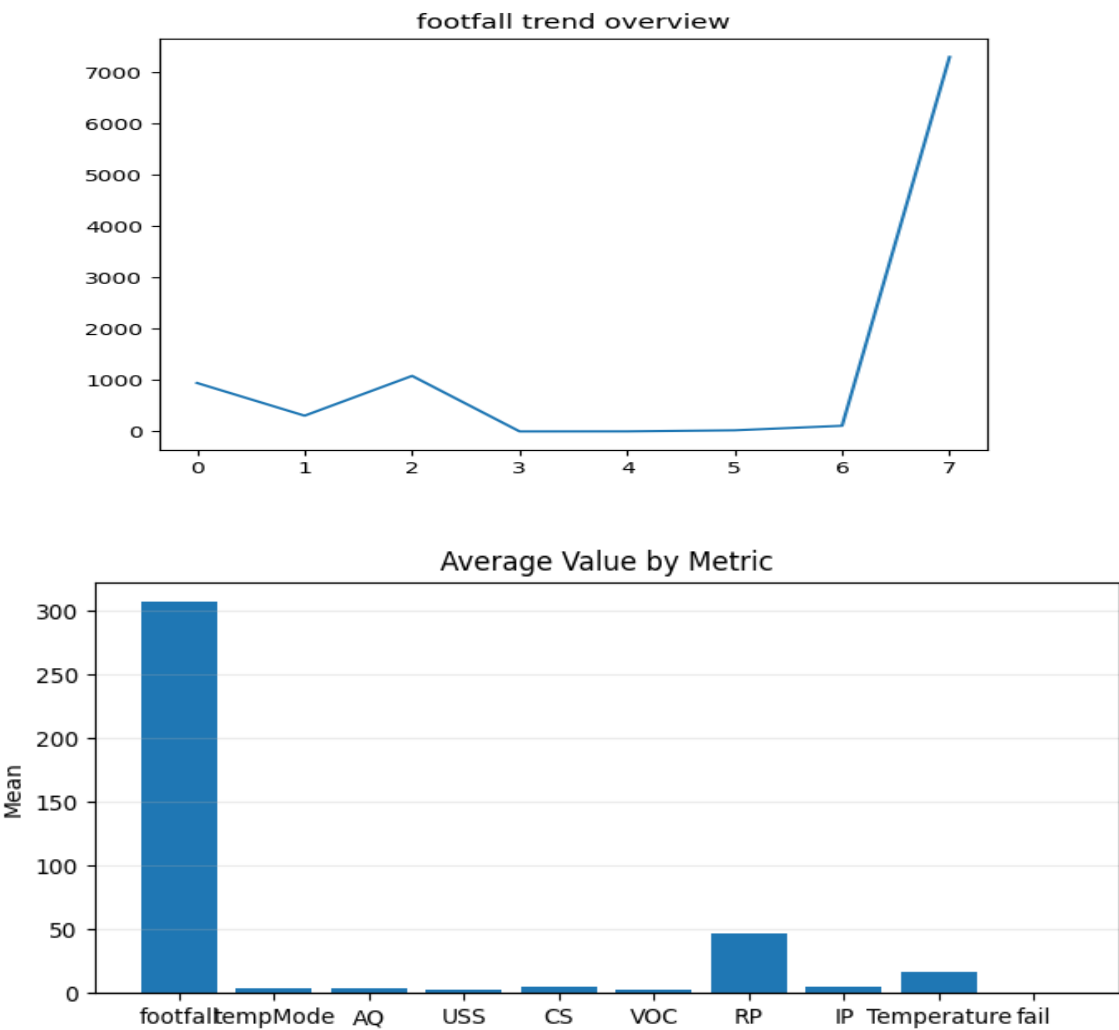


Chart Interpretation:

- footfall: bar chart centers near mean 306.381; line trend indicates down. Mean/median relation suggests a right-skewed distribution.
- tempMode: bar chart centers near mean 3.728; line trend indicates up. Mean/median relation suggests a right-skewed distribution.
- AQ: bar chart centers near mean 4.325; line trend indicates down. Mean/median relation suggests a right-skewed distribution.
- USS: bar chart centers near mean 2.940; line trend indicates up. Mean/median relation suggests a left-skewed distribution.
- CS: bar chart centers near mean 5.394; line trend indicates down. Mean/median relation suggests a left-skewed distribution.

- VOC: bar chart centers near mean 2.842; line trend indicates up. Mean/median relation suggests a right-skewed distribution.
- RP: bar chart centers near mean 47.043; line trend indicates up. Mean/median relation suggests a right-skewed distribution.
- IP: bar chart centers near mean 4.566; line trend indicates down. Mean/median relation suggests a right-skewed distribution.
- Temperature: bar chart centers near mean 16.332; line trend indicates up. Mean/median relation suggests a left-skewed distribution.
- fail: bar chart centers near mean 0.416; line trend indicates stable. Mean/median relation suggests a right-skewed distribution.
- Histogram bins show where value density is concentrated between quartile ranges.
- Pie chart summarizes stable vs attention-needed metrics from focus-area rules.

## Data Change Explanation

### Data Change Analysis:

- footfall: value range 0.000 to 7300.000 (spread 7300.000), middle 50% spans 109.000. Trend is down; volatility appears high; focus area is 18 anomalies, high variance.
- tempMode: value range 0.000 to 7.000 (spread 7.000), middle 50% spans 6.000. Trend is up; volatility appears high; focus area is high variance.
- AQ: value range 1.000 to 7.000 (spread 6.000), middle 50% spans 3.000. Trend is down; volatility appears high; focus area is high variance.
- USS: value range 1.000 to 7.000 (spread 6.000), middle 50% spans 2.000. Trend is up; volatility appears high; focus area is high variance.
- CS: value range 1.000 to 7.000 (spread 6.000), middle 50% spans 1.000. Trend is down; volatility appears high; focus area is 13 anomalies, high variance.
- VOC: value range 0.000 to 6.000 (spread 6.000), middle 50% spans 4.000. Trend is up; volatility appears high; focus area is high variance.
- RP: value range 19.000 to 91.000 (spread 72.000), middle 50% spans 24.000. Trend is up; volatility appears high; focus area is high variance.
- IP: value range 1.000 to 7.000 (spread 6.000), middle 50% spans 3.000. Trend is down; volatility appears high; focus area is high variance.
- Temperature: value range 1.000 to 24.000 (spread 23.000), middle 50% spans 7.000. Trend is up; volatility appears high; focus area is high variance.
- fail: value range 0.000 to 1.000 (spread 1.000), middle 50% spans 1.000. Trend is stable; volatility appears high; focus area is high variance.

## Key Findings

- footfall: values range from 0.00 to 7300.00, average around 306.38. Trend appears down; review note: 18 anomalies, high variance.
- tempMode: values range from 0.00 to 7.00, average around 3.73. Trend appears up; review note: high variance.
- AQ: values range from 1.00 to 7.00, average around 4.33. Trend appears down; review note: high variance.
- USS: values range from 1.00 to 7.00, average around 2.94. Trend appears up; review note: high variance.
- CS: values range from 1.00 to 7.00, average around 5.39. Trend appears down; review note: 13 anomalies, high variance.
- VOC: values range from 0.00 to 6.00, average around 2.84. Trend appears up; review note: high variance.

## AI-Powered Insights

**Executive Summary:** This report explains the analysis results for data.csv in clear language. The findings are based on the same data used in the website dashboard and do not introduce different calculations.

**Data Overview:** The dataset includes approximately 944 records and 10 variables. The analysis focused on trend behavior, consistency, unusual values, and confidence indicators.

**Key Findings:** Main focus areas include footfall:18 anomalies, high variance, tempMode:high variance, AQ:high variance, USS:high variance, CS:13 anomalies, high variance, VOC:high variance, RP:high variance, IP:high variance, Temperature:high variance, fail:high variance. Trend signals show footfall:down, tempMode:up, AQ:down, USS:up, CS:down, VOC:up, RP:up, IP:down, Temperature:up, fail:stable.

**AI-Powered Insights:** The observed changes may reflect shifts in operating conditions, collection methods, or natural process variation. Results should be reviewed with context from the experiment timeline.

**Confidence & Reliability:** The analysis is generally reliable for decision support, but confidence can be reduced when unusual values or missing entries are present.

## Technical Review (Advanced)

Technical Review (Advanced):

- footfall: mean=306.3814, std=1082.6067, min=0.0000, max=7300.0000, IQR=109.0000, CV=3.5335, trend=down, focus=18 anomalies, high variance, confidence=0.98.
- tempMode: mean=3.7278, std=2.6772, min=0.0000, max=7.0000, IQR=6.0000, CV=0.7182, trend=up, focus=high variance, confidence=1.0.
- AQ: mean=4.3252, std=1.4384, min=1.0000, max=7.0000, IQR=3.0000, CV=0.3326, trend=down, focus=high variance, confidence=1.0.
- USS: mean=2.9396, std=1.3837, min=1.0000, max=7.0000, IQR=2.0000, CV=0.4707, trend=up, focus=high variance, confidence=1.0.
- CS: mean=5.3941, std=1.2693, min=1.0000, max=7.0000, IQR=1.0000, CV=0.2353, trend=down, focus=13 anomalies, high variance, confidence=0.99.
- VOC: mean=2.8422, std=2.2733, min=0.0000, max=6.0000, IQR=4.0000, CV=0.7999, trend=up, focus=high variance, confidence=1.0.
- RP: mean=47.0434, std=16.4231, min=19.0000, max=91.0000, IQR=24.0000, CV=0.3491, trend=up, focus=high variance, confidence=1.0.
- IP: mean=4.5657, std=1.5993, min=1.0000, max=7.0000, IQR=3.0000, CV=0.3503, trend=down, focus=high variance, confidence=1.0.
- Temperature: mean=16.3316, std=5.9748, min=1.0000, max=24.0000, IQR=7.0000, CV=0.3658, trend=up, focus=high variance, confidence=1.0.
- fail: mean=0.4163, std=0.4932, min=0.0000, max=1.0000, IQR=1.0000, CV=1.1847, trend=stable, focus=high variance, confidence=1.0.

## Technical Recommendations (Advanced)

Technical Recommendations (Advanced):

- footfall: high coefficient of variation (CV=3.5335); apply robust thresholding and recalibration checks.
- footfall: anomaly-heavy profile detected; run outlier root-cause analysis against raw capture logs.
- footfall: persistent downward trend; validate process degradation assumptions and lower tolerance thresholds.
- tempMode: high coefficient of variation (CV=0.7182); apply robust thresholding and recalibration checks.
- tempMode: persistent upward trend; evaluate drift alarms and upper tolerance boundaries.

- AQ: high coefficient of variation (CV=0.3326); apply robust thresholding and recalibration checks.
- AQ: persistent downward trend; validate process degradation assumptions and lower tolerance thresholds.
- USS: high coefficient of variation (CV=0.4707); apply robust thresholding and recalibration checks.
- USS: persistent upward trend; evaluate drift alarms and upper tolerance boundaries.
- CS: anomaly-heavy profile detected; run outlier root-cause analysis against raw capture logs.
- CS: persistent downward trend; validate process degradation assumptions and lower tolerance thresholds.
- VOC: high coefficient of variation (CV=0.7999); apply robust thresholding and recalibration checks.

## Special AI Recommendations

Priority actions below are generated from trend volatility, confidence, and focus-area signals.

Abstract: The analysis processed 944 rows across 10 columns to evaluate quality, trend behavior, and operational risk.

Introduction: Recommendations are mapped to Performance (25%), Innovation (25%), Operational Impact (25%), and Responsible AI (25%).

Materials and Methods: Data quality, distribution spread, quartiles, trend direction, focus-area tags, and confidence values were evaluated per metric.

Results:

- footfall: mean=306.3814, std=1082.6067, range=0.0000..7300.0000, IQR=109.0000, CV=3.534, trend=down, focus=18 anomalies, high variance, confidence=0.98.
- tempMode: mean=3.7278, std=2.6772, range=0.0000..7.0000, IQR=6.0000, CV=0.718, trend=up, focus=high variance, confidence=1.0.
- AQ: mean=4.3252, std=1.4384, range=1.0000..7.0000, IQR=3.0000, CV=0.333, trend=down, focus=high variance, confidence=1.0.
- USS: mean=2.9396, std=1.3837, range=1.0000..7.0000, IQR=2.0000, CV=0.471, trend=up, focus=high variance, confidence=1.0.
- CS: mean=5.3941, std=1.2693, range=1.0000..7.0000, IQR=1.0000, CV=0.235, trend=down, focus=13 anomalies, high variance, confidence=0.99.
- VOC: mean=2.8422, std=2.2733, range=0.0000..6.0000, IQR=4.0000, CV=0.800, trend=up, focus=high variance, confidence=1.0.
- RP: mean=47.0434, std=16.4231, range=19.0000..91.0000, IQR=24.0000, CV=0.349, trend=up, focus=high variance, confidence=1.0.
- IP: mean=4.5657, std=1.5993, range=1.0000..7.0000, IQR=3.0000, CV=0.350, trend=down, focus=high variance, confidence=1.0.
- Temperature: mean=16.3316, std=5.9748, range=1.0000..24.0000, IQR=7.0000, CV=0.366, trend=up, focus=high variance, confidence=1.0.
- fail: mean=0.4163, std=0.4932, range=0.0000..1.0000, IQR=1.0000, CV=1.185, trend=stable, focus=high variance, confidence=1.0.

Discussion:

- Risk stratification: high risk=9, medium risk=1, low-confidence=0 metrics.
- High-priority metrics: footfall, tempMode, AQ, USS, CS, VOC, RP, IP, Temperature.
- Medium-priority metrics: fail.
- Recommendation confidence improves when calibration records, sampling context, and anomaly root-cause notes are attached to each run.

Conclusion: The dataset supports decision guidance when recommendations are executed with metric-level evidence, uncertainty disclosure, and safety checks.

Recommended Next Actions (criteria-aligned):

- 1) [Performance 25%] footfall: anomaly concentration detected; verify raw acquisition logs, sensor drift, and boundary thresholds.

- 2) [Innovation 25%] footfall: high variability (CV=3.534); add adaptive control limits and confidence-banded recommendations.
- 3) [Performance 25%] footfall: downward trend detected; verify if decline matches expected process behavior and acceptable bounds.
- 4) [Innovation 25%] tempMode: high variability (CV=0.718); add adaptive control limits and confidence-banded recommendations.
- 5) [Performance 25%] tempMode: upward trend detected; review upper tolerance limits and likely drift drivers.
- 6) [Innovation 25%] AQ: high variability (CV=0.333); add adaptive control limits and confidence-banded recommendations.
- 7) [Performance 25%] AQ: downward trend detected; verify if decline matches expected process behavior and acceptable bounds.
- 8) [Innovation 25%] USS: high variability (CV=0.471); add adaptive control limits and confidence-banded recommendations.
- 9) [Performance 25%] USS: upward trend detected; review upper tolerance limits and likely drift drivers.
- 10) [Performance 25%] CS: anomaly concentration detected; verify raw acquisition logs, sensor drift, and boundary thresholds.
- 11) [Operational Impact 25%] Log metric-level evidence, confidence tags, and safety outcomes in a consistent audit trail for transparent reporting.

## Recommendations

Detailed Recommendations:

Priority 1 - Immediate Monitoring:

- footfall: focus=18 anomalies, high variance, trend=down, CV=3.534, confidence=0.98. Prioritize calibration, anomaly review, and completeness checks before downstream decisions.
- tempMode: focus=high variance, trend=up, CV=0.718, confidence=1.0. Prioritize calibration, anomaly review, and completeness checks before downstream decisions.
- AQ: focus=high variance, trend=down, CV=0.333, confidence=1.0. Prioritize calibration, anomaly review, and completeness checks before downstream decisions.
- USS: focus=high variance, trend=up, CV=0.471, confidence=1.0. Prioritize calibration, anomaly review, and completeness checks before downstream decisions.
- CS: focus=13 anomalies, high variance, trend=down, CV=0.235, confidence=0.99. Prioritize calibration, anomaly review, and completeness checks before downstream decisions.
- VOC: focus=high variance, trend=up, CV=0.800, confidence=1.0. Prioritize calibration, anomaly review, and completeness checks before downstream decisions.

Priority 2 - Investigation Actions:

- Compare unusual spikes and drops against experiment logs, equipment events, and environment changes from the same time window.
- Validate whether trend changes are expected process behavior or indicate drift needing correction.

Priority 3 - Improvement Actions:

- Standardize data collection checkpoints (calibration, timestamp validation, completeness checks) before each run.
- Add weekly review of confidence and focus-area flags to improve reliability over time.

Priority 4 - Governance and Safety:

- Require human approval for decisions when confidence is low or quality warnings are present.
- Keep a decision log linking each action to the supporting metric evidence and confidence value.

## Priority Recommendation Guide



| Priority Level     | Meaning                          | Recommended Action                  |
|--------------------|----------------------------------|-------------------------------------|
| High Priority      | Immediate attention required     | Priority 1 - Immediate Monitoring:  |
| Moderate Attention | Track closely and follow up      | Priority 2 - Investigation Actions: |
| Low Concern        | Routine monitoring is sufficient | Priority 3 - Improvement Actions:   |

Confidence & Reliability

The analysis is generally reliable when interpreted with the same context shown in the website dashboard. Attention is recommended for: footfall, tempMode, AQ, USS, CS, VOC, RP, IP, Temperature, fail due to unusual patterns or data quality concerns. Missing values or unusual spikes can reduce certainty, so regular monitoring and data quality checks are advised.

Document Notes

- Document Context:
- Source file: data.csv
  - File type: csv
  - Structural profile: rows=944, columns=10
  - This report is generated for a general lab-review workflow and operational decision support.

Legal & Compliance Context

The report automatically adapts reference guidance to detected data context. Domain tags: general.

- Apply recommendations with human oversight and retain an auditable decision log linking actions to evidence.
- Confirm data minimization, access control, and retention limits before operational use.

AI Responsibility & Transparency

This report is generated using AI technologies provided through Microsoft Azure. Recommendations are based on available data and are intended to support decision-making. Validate critical decisions with domain experts.

Appendix

Additional summary details are provided below for reference.

| Metric   | Mean     | Std       | Min    | Max       | Trend | Focus Area                  | Confidence |
|----------|----------|-----------|--------|-----------|-------|-----------------------------|------------|
| footfall | 306.3814 | 1082.6067 | 0.0000 | 7300.0000 | down  | 18 anomalies, high variance | 0.98       |
| tempMode | 3.7278   | 2.6772    | 0.0000 | 7.0000    | up    | high variance               | 1.0        |
| AQ       | 4.3252   | 1.4384    | 1.0000 | 7.0000    | down  | high variance               | 1.0        |
| USS      | 2.9396   | 1.3837    | 1.0000 | 7.0000    | up    | high variance               | 1.0        |

| Metric      | Mean    | Std     | Min     | Max     | Trend  | Focus Area                  | Confidence |
|-------------|---------|---------|---------|---------|--------|-----------------------------|------------|
| CS          | 5.3941  | 1.2693  | 1.0000  | 7.0000  | down   | 13 anomalies, high variance | 0.99       |
| VOC         | 2.8422  | 2.2733  | 0.0000  | 6.0000  | up     | high variance               | 1.0        |
| RP          | 47.0434 | 16.4231 | 19.0000 | 91.0000 | up     | high variance               | 1.0        |
| IP          | 4.5657  | 1.5993  | 1.0000  | 7.0000  | down   | high variance               | 1.0        |
| Temperature | 16.3316 | 5.9748  | 1.0000  | 24.0000 | up     | high variance               | 1.0        |
| fail        | 0.4163  | 0.4932  | 0.0000  | 1.0000  | stable | high variance               | 1.0        |

Detailed Data Table

| Metric      | Count | Mean     | Std       | Min  | Max    |
|-------------|-------|----------|-----------|------|--------|
| footfall    | 944.0 | 306.3814 | 1082.6067 | 0.0  | 7300.0 |
| tempMode    | 944.0 | 3.7278   | 2.6772    | 0.0  | 7.0    |
| AQ          | 944.0 | 4.3252   | 1.4384    | 1.0  | 7.0    |
| USS         | 944.0 | 2.9396   | 1.3837    | 1.0  | 7.0    |
| CS          | 944.0 | 5.3941   | 1.2693    | 1.0  | 7.0    |
| VOC         | 944.0 | 2.8422   | 2.2733    | 0.0  | 6.0    |
| RP          | 944.0 | 47.0434  | 16.4231   | 19.0 | 91.0   |
| IP          | 944.0 | 4.5657   | 1.5993    | 1.0  | 7.0    |
| Temperature | 944.0 | 16.3316  | 5.9748    | 1.0  | 24.0   |
| fail        | 944.0 | 0.4163   | 0.4932    | 0.0  | 1.0    |

References (AI Use)

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